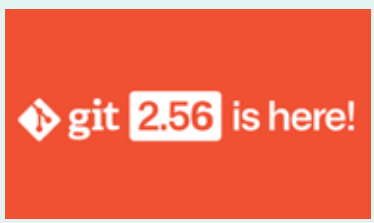
Git 2.56 improves development workflows

The open source Git project has released Git 2.56, bringing new features and performance improvements developed by more than 104 contributors, including 39 first-time contributors. The release focuses on making common version-control workflows safer while improving Git's ability to handle large repositories.

One of the notable additions is 'git add –resolved', which provides a more targeted way to stage files after resolving merge conflicts. Unlike broader staging commands, it considers only paths that are currently unmerged. Git also checks for remaining conflict markers before staging regular files, helping prevent unresolved conflicts from being accidentally committed.

Git 2.56 improves merge-base calculation, allowing Git to stop searching once it determines that no additional common ancestors can exist. This can significantly reduce the amount of history that needs to be traversed in large repositories. The release also makes path-walk repacking more practical for repository hosts. This method groups objects based on their location in a repository's tree and can produce substantially smaller pack files.

Other changes include new capabilities for the experimental git history and 'git refs' commands, bulk deletion of merged branches, automatic cleanup after 'git bisect run', improved path tracking with 'git log –follow', and better handling of partial-clone objects.



Linus Torvalds credits Claude Code for Linux kernel optimisation

Linux creator Linus Torvalds has taken his use of artificial intelligence a step further, leveraging Anthropic's Claude Code assistant to help author new performance-focused kernel modifications.

While preparing for the Linux 7.3-rc7 release, observers noticed a rare new branch appear in Torvalds' primary repository: noref-filemap-read-with-llm. The experimental code targets a longstanding Linux bottleneck where concurrent small reads on a single file create contention around folio reference counting. By enabling page cache reads without folio references, the patch series achieves significant performance gains under heavy multi-threaded workloads.

In a rare move for mainstream kernel development, Torvalds explicitly acknowledged the role of AI in the patch series' main cover letter: "The code, the test programs, the review behind the list above and the first draft of this text were done with an LLM coding assistant (Claude Code) under my direction. The measurements were run on my machine."

Initially, he did not rate AI highly, but he has since changed his mind. Indeed, he has been quoted saying, "I actually really like using AI."

Meta launches Muse Gadgets for DIY hardware

Meta has introduced Muse Gadgets, an open source hardware initiative that allows developers and hobbyists to build custom devices connected to its Muse personal AI agent. The project provides an ESP32 Device SDK and Linux Device SDK, enabling makers to use off-the-shelf development boards and connect Muse to screens, buttons, audio hardware, sensors and actuators. The SDKs and firmware are released under the Apache 2.0 licence.



The ESP32 SDK is designed for boards makers can program and customise, with support for displays, audio input and output, and additional sensors. Meta currently showcases several compatible devices, including ESP32-based boards with touchscreens and displays. Developers can also build support for other boards, allowing the project to extend beyond the hardware featured by Meta.

For Linux-based hardware, the Linux Device SDK can turn a Raspberry Pi or another Linux computer into a Muse gadget. This allows developers to connect Muse with local applications and services, including home assistant setups.

The project gives the maker community access to the source code for the device SDKs and firmware, with Meta encouraging developers to modify the software, build new hardware integrations and contribute improvements.

Firefox gets a new design

Mozilla has released Firefox 157, bringing a redesigned user interface to its open source web browser. The release expands the Project Nova redesign, with changes to navigation, tabs, icons, spacing and browser surfaces intended to modernise the browsing experience while retaining Firefox's focus on customisation.

The redesign changes several core parts of the browser interface, including the toolbar, menus, tabs and settings. Mozilla has also introduced updated themes, wallpapers and other personalisation options. The changes have been developed by the Firefox community, with Mozilla using feedback from users and testing through its Nightly development channel before wider deployment.